COURSE 2 · SECTION 8 · EXERCISE

Change cycle time and backlog

Section 8 (videos S08.1 and S08.2) · Engineering change management

Start: Section09_1_Start.pbix or your Engxtech_S08_2.pbix; save a copy as Engxtech_S08_Exercise.pbix

Time: about 25 min

Data: the course data folder

DESIGN QUESTION

How long does an engineering change take from request to close, how big is the open backlog at a given date, what do changes cost, and how did ECR-3200 compare with a typical change?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

Every change keeps its ECR number for life: an ECR (request) becomes an ECO (order) once approved. ParentChangeID is filled when a change reworks an earlier one. CostImpact_USD is negative for cost reductions. The video added CycleDays and the four stage-day columns to Changes, and the cycle, backlog, rework and cost measures to _Measures.

Your tasks

1. In your exercise copy, read the Changes page cards: **Median Cycle Days** and **P90 Cycle Days**, and the stage medians (Median Approval Days).
2. Hover the **Open Backlog** line by Dates[MonthStart]: read Dec 2025 and Sep 2026.
3. Read the cards **Rework Changes**, **Net Change Cost**, **Added Change Cost** and **Change Savings**. Label the net figure as net.
4. New table visual: Changes[ChangeID] filtered to ECR-3200 with RequestDate, TriageDate, ApprovalDate, ImplementDate, CloseDate, CycleDays and ApprovalDays (Don't summarize). Compare with the median and p90.
5. New measure in _Measures: **Changes Slower Than ECR-3200**: closed changes with CycleDays above 123. How rare was ECR-3200?

Check your answer

Check	Expected
Median closed cycle time	**47.0 d**
P90 closed cycle time	**83.7 d**
Median approval stage	**13.0 d**
Open backlog as of 31 Dec 2025	**15**
Open backlog as of 30 Sep 2026	**24**
Rework changes	**9**
Net / added / savings (USD)	**158,004 / 251,790 / -93,786**
ECR-3200 cycle / approval stage	**123 d / 57 d**
Closed changes slower than ECR-3200	**1**

Hint. The backlog measure ignores the Dates filter (REMOVEFILTERS) and then applies its own rule: requested on or before the last date in the filter, and not closed by that date.

Stuck? The worked solution with the full DAX and M is in Solutions/S08_Solution.pdf.